

Time synchronization

Visual guide and experimental report

Wireless IR motion capture | Four ESP32-S3 nodes

Experiments: 8-9 September 2026

Best-supported choice: live FTM + MCPWM

1.688 microseconds: largest observed error in the new FTM trial.

Limited repetitions; results measure GPIO phase, not exposure.

Reading guide

- 2** Comparison of the five clock methods
- 3-4** Simple explanation and timing diagrams
- 5** Setup photograph and hardware identification
- 6** Logic screenshot and recorded edge detail
- 7** Comparison of the six output backends
- 8 onward** Full methods, results, failures, limitations and references

Includes the visual supplement and the detailed report in one document.

Nine of fifteen planned new 60-second comparison trials completed.

Acquisition failures and unequal repetition counts remain explicit.

The analyzer identification is clarified using the supplied photograph.